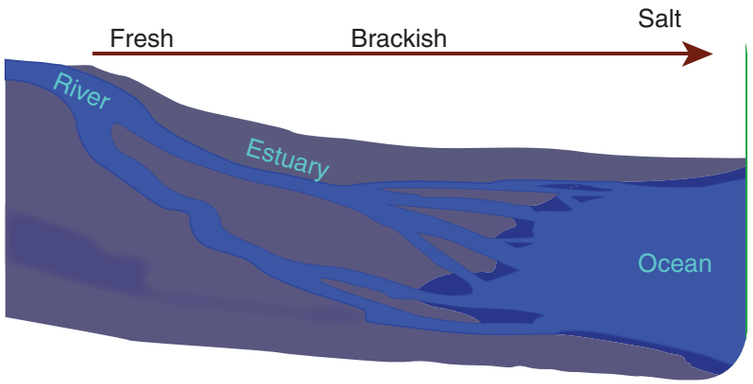


Figure 9.4: Estuary

fowl fecal matter deposited on the ground. Sometimes overflow from sewage systems contributes to these higher number of bacteria in the water. Microbiota are the primary producers in the aquatic ecosystem since they play a major role in food chain in water. Drifting microbial life of aquatic environment is called plankton. It is composed of phytoplankton and zooplankton. The bottom region of the water harbours largest number of microorganisms called as benthic microorganisms. Microorganisms have been found at all depths and at all latitudes within the ocean. Microorganisms are abundant near shore regions, where organic pollution as well as microbial pollution from the surrounding environment occurs.

Algae are common in these environments. Because of their dependence on light as a source of energy, algae occur primarily in the upper strata of the oceans and seas. Although they constitute a vital part of the food chain in the marine environments, they also can be a nuisance and threat to other forms of life. Nutrient levels in such environments can significantly increase as a result of sewage plant discharges. Under such conditions algal blooms are common.



Barophiles are bacteria that thrive deep in the ocean and live at pressures of 400-700 atmosphere but die at 1 atmosphere. A strain MT-41, a bacteria is isolated from the marine sediments in the Mariana Trench near the Phillippines where the depth exceeds 10 kilometres. This strain has optimum growth at a pressure of 709 bar.

9.3.3 Freshwater Microflora

The study of fresh water ecosystem is referred to as Limnology. The major bodies of fresh water are rivers, streams, swamps, marshes and lakes. The fresh water ecosystem is divided into lentic (still water) and lotic (flowing water) ecosystems. Lentic ecosystem is divided into zones based on light penetration and temperature. They are littoral, limnetic and profundal zones (Figure 9.5).

Most lakes are surrounded by rooted vegetation in a large littoral zone along the shore. Light penetrates the shallow littoral and open-water limnetic zone but is unable to reach the profundal zone in the deep portions of many lakes.

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